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Notion of a good death for patients with cancer: A qualitative systematic review protocol

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Notion of a good death for patients with cancer: A qualitative systematic review protocol

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Abstract

Introduction: Promoting the quality of death and achieving a 'good death' are among the most important goals of palliative care in the area of cancer care. Little research is concentrated on patients with cancer to synthesise their view on the notion of a good death and its associated factors. By eliciting the core elements of a good death, the review aims to identify potential unmet needs of patients with cancer and reveal their common values and care preferences at end of life. The review also has the potential to inform the development of guidelines for clinical care and shared decision making in palliative care practice.

Methods and analysis: The qualitative systematic review will be conducted and reported according to the Joanna Briggs Institute (JBI) methodology for systematic review and synthesis of qualitative data and the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols. Comprehensive search will be performed in six English and two Chinese databases from inception to 30 October 2021 for the retrieval of relevant qualitative articles focusing on the notion of a good death from the perspective of patients with cancer. Data will be extracted using the JBI standardised data extraction tool for qualitative research. The quality of the included studies will be critically appraised using the JBI Qualitative Assessment and Review Instrument. Data extraction and appraisal will be conducted by two reviewers independently. Meta-aggregative approach and narrative summary will be used in synthesising data.

Ethics and dissemination: Ethical approval is not needed. We will disseminate the

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findings through international conference related to cancer care or palliative care. The final review will be submitted and published in a peer-reviewed journal.

Keywords: good death, quality of dying, cancer, synthesis, review

Strengths and Limitations of this study

- The qualitative systematic review will be conducted according to the Joanna Briggs Institute methodology for systematic review and synthesis of qualitative data for a high level of rigor.
- The review is designed to enhance understanding of a good death and facilitate the building of an empirical intervention model that promotes 'good death'.
- The synthesised findings are expected to have a global impact in the area of cancer care given that the review will include patients with different types of cancer in different cultural contexts.
- Some studies can be overlooked because this review will only include studies published in English and Chinese.

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Introduction

The International Agency for Research on Cancer reported 19.3 million new cases of cancer and 10.0 million deaths in 2020. Although the survivorship of a number of cancer types has been substantially prolonged because of the advances in medicine and technology over the recent decades, cancer is still frequently considered synonymous with the term 'death', especially advanced cancer. A cross-sectional study of 402 cancer deaths in Canada reported that only 39% of deaths were rated 'good' or 'almost perfect' [1]. Promoting the quality of death and achieving a 'good death' is one of the most important goal of palliative care and care of cancer.

Yet what is a good death? Previous research has shown that the concept is complex. The concept of good death states that patients with cancer and stakeholders, including families and caregivers, can be free from distress and suffering at end-of-life and dying in accordance with patients' wishes and clinical, cultural and ethical standards [2]. A qualitative research on patients with cancer, families and healthcare providers in Japan showed that freedom from pain or physical and psychological symptoms is the most frequently cited attribute of a good death, followed by having good family relationship, dying in the preferred place and having good relationship with medical staff [3]. Another study on the bereaved families of patients with cancer reported that physical and psychological suffering and caregivers' distress and emotions are important factors associated with 'good death' [4].

Proxy views on quality of death may be primarily based on the caregivers' perspective, and patients may have different perceptions. Some studies showed that

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good death is dependent on the perspectives of dying individuals [5 6]. A 2007 Sweden qualitative study with 66 adult palliative patients with cancer indicated that patients viewed death as a process rather than an event [7]. A good death was deemed a 'successful dying process' when patients are living with the prospect of imminent death, preparing for death and dying comfortably. However, patients in Eastern countries (e.g. China) have faced challenges when preparing for death because of lower frequencies of bad news (e.g. diagnosis and prognosis) breaking practices and stricter taboos on death discussions than those in Western countries. They are more likely to miss optimal time to accept impending death and make preparations for definite departure [8]. Hence, understanding how patients from different cultures view a good death can help them make good preparations for imminent death through an acculturation approach.

Several reviews have been conducted to enhance understanding of the notion of a good death. However, the implications of the reviews cannot be generalised and used in promoting a good death in cancer because of their limitations. A recent review summarised the notion of a good death from patients' perspectives [9]. Control of pain and symptom, clear decision-making, feeling of closure, being seen and perceived as a person, preparation for death and being still able to give something to others were summarised as core elements for a 'good death'. However, the review included a heterogeneous group of patients with various advanced diseases, such as acquired immune deficiency syndrome, cardiovascular disease, chronic obstructive pulmonary disease and cancer. The conclusions may not perfectly fit patients with cancer. Other reviews included patients with multiple stakeholders, such as families and healthcare

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providers, or a combination of patients with cancer and non-malignant disease [10 11]. More concentrated attention on patients' perspective helps elicit and meet their needs in the end of life.

Exploring the core components of a 'good death' from the perspective of patients with cancer and factors associated with their experience is imperative given the growth of palliative care over the last decade and the increased awareness of the right-to-die movement. To the best of our knowledge, no review has synthesised qualitative studies to reveal the notion of a 'good death' specifically from the perspective of patients with cancer. By eliciting the core elements of a good death, the review aims to identify potential unmet needs of patients with cancer and reveal their common values and care preferences at end of life. The review also has the potential to inform the development of guidelines for clinical care and shared decision making in palliative care practice.

Methods

Aim

This review aims to identify, appraise and synthesise the existing qualitative evidence exploring the notion of good death of patients with cancer. The review will particularly seek to answer the following questions:

1. What are the characteristics of the existing studies that explored a good death from the perspective of patients with cancer?
2. What are the core elements of a good death for patients with cancer?
3. What are the main factors associated with a good death?

Study design

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A protocol was developed according to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) and the Joanna Briggs Institute (JBI) methodology for systematic review and synthesis of qualitative data [12]. Synthesis of qualitative research will be used because we aim to present a comprehensive understanding of patients' perceptions on a 'good death' [13]. The review protocol was submitted to the International prospective register of systematic reviews (PROSPERO) for registration and now is under review.

Inclusion criteria

Types of participants: This review will consider studies that only included adult patients (over 18 years old) with cancer. Studies involving families, healthcare providers, experts or social workers or a mixed group of patients with cancer and other participants (e.g. patients diagnosis with other diseases, families, healthcare providers and other stakeholders) will be excluded.

Phenomena of interest: This review will consider studies that explore the notion of good death (e.g. the views/perspectives/attitudes/belief about a good death) or main factors (environmental, illness-related or other possible factors) that associated with a good death. Epidemiological studies or studies focusing on certificate of death will be excluded.

Context: This review will include studies conducted in any setting where patients with cancer can be cared for, including but not limited to hospitals, hospices, long-term care facilities and private residences of patients.

Types of studies: Eligible studies should employ a qualitative methodology

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including but not limited to phenomenology, ethnography and grounded theory. Quantitative studies, mixed-method studies, case studies, reviews, commentaries, letters, opinion papers and conference abstracts will be excluded. Only studies published in English and Chinese language will be included.

Search methods

This review will consider papers published by academic publishers, as well as grey literature. A comprehensive search will firstly be performed on MEDLINE (via PubMed), Embase, PsychINFO, CINAHL, Scopus, Web of Science and two Chinese databases (i.e. CNKI and WANGFANG DATA) from inception to October 2021. Grey literature search will be performed via Google Scholar with the keywords. The initial search strategy is showed in table 1. Reference lists of studies that meet the inclusion criteria will be screened for additional studies.

Table 1 The initial search strategy

‘cancer’ OR ‘neoplasms’
AND
‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’

The initial search strategies were carried out in PubMed in July 2021, and the following queries were used: ("cancer"[Title/Abstract] OR "neoplasms"[Title/Abstract]) AND ("good death"[Title/Abstract] OR "quality of death"[Title/Abstract] OR "quality of dying"[Title/Abstract] OR "good dying"[Title/Abstract]). Two-hundred and sixty-four results were identified. The first author will screen these results firstly with the purpose of refining and developing specific search strategies for each database.

Study selection

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References identified in the aforementioned databases will be imported into EndNote 20. In the first step, duplicates will be removed with EndNote's deduplicating capabilities, followed by manual screening. Second, study screening will be performed by two reviewers independently, and Covidence will be used, which is a web-based platform that streamlines the implementation of systematic reviews. After title and abstract screening, full texts will be checked according to the inclusion criteria by two reviewers independently. All reasons for exclusion will be recorded, and the results of screening will be cross-checked between the two reviewers. Any disagreements will be resolved through discussion with the third reviewer. In the third step, all the reference lists in the included studies will be checked for additional studies.

Data extraction

Data extraction will be conducted using the JBI standardised data extraction tool for qualitative research from JBI SUMARI by two reviewers independently. The data to be extracted will include specific details about the authors and year of publication, journals in which the studies were published, methodologies, methods, phenomena of interest, settings and contexts, characteristics of the participants (age, gender, type of cancer and illness-related information), data analysis, primary findings, authors' conclusions and comments [14]. Any disagreement between the two reviewers will be resolved by the third reviewer.

Quality appraisal

Included studies will be critically appraised independently using the JBI Qualitative Assessment and Review Instrument (JBI QARI) and will be conducted in

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JBI System for the Unified Management, Assessment and Review of Information (JBI SUMARI) [15]. JBI SUMARI is a software that supports the entire systematic review process (e.g. quality appraisal, data extraction and data synthesis) for different types of systematic reviews. The instrument consists of 10 items: five items appraising the congruities between research methodology and stated philosophical perspective, research question, data collection methods and data analysis; two items appraising the researchers' roles and cultures; one item appraising the representation of participants and their voices; one item for ethical approval; and one item for the interpretation of the data [12]. Two reviewers will be blinded to each other's assessment. The result of appraisal will only be compared after the completion of the initial appraisal of an article. In case of any disagreement, a third reviewer will assist in the appraisal, discussion of the study quality and consensus.

Data synthesis

Meta-aggregative approach will be adopted for the synthesis of the findings. This approach is aligned with the philosophy of pragmatism, suitably answers a specific question and summarises views on good death [14]. First, the findings of the qualitative research will be assembled. Then, the findings will be categorised into groups with different codes according to similarity in meaning. Lastly, the categories will be aggregated for the generation of a set of synthesis findings, which can be used as a basis for evidence-based practice[16]. Only the unequivocal and credible findings will be included in the aggregation. Non-supported findings will be presented separately. If the textual pooling is not possible, the findings will be presented in a narrative form.

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Confidence in the synthesised findings of the review

The confidence in the synthesised findings of the review will be assessed in accordance with the ConQual, and dependability and credibility will be determined [17]. In ConQual, the dependability of findings is evaluated using five questions assessing the congruities in the JBI QARI. The credibility is established by assessing the congruency between the authors' interpretation and the supporting data. The ConQual will then rank and assign the confidence of the qualitatively synthesised findings into four levels (i.e. High, Moderate, Low or Very low) through comprehensively considering the dependability and credibility assessment. A ConQual summary will be presented along with the type of study, dependency score, credibility score and overall ConQual rank.

Ethical considerations

Ethical approval is not applicable for this protocol nor the systematic review of qualitative studies. After the completion of the review, the findings will be shared through presentation at international conference and in an international peer-reviewed journal.

Discussion

This will be the first systematic review of qualitative studies that synthesises the findings of a good death specifically from the perspective of patients with cancer. The review is designed to enhance understanding of a good death and has the potential to support the development of an empirical model in promoting a 'good death'. Patients' view on a good death and factors associated with their experiences will be summarised,

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and thus a new knowledge base will be added and detailed implications will be provided in the area of cancer care. The synthesised findings are expected to have a global impact in the area of cancer care given that the review will include patients with different types of cancer and cultural contexts.

Author Contributors

XIAO Jinnan designed the study and drafted the manuscript. HUANG Chongmei supervised the study design and critically revised the manuscript. DING Jinfeng contributed to the methods and critically revised the manuscript.

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Competing interest

None declared

Patient consent for publication

Not required

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References:

1. Hales S, Chiu A, Husain A, et al. The quality of dying and death in cancer and its relationship to palliative care and place of death. *J Pain Symptom Manage* 2014;48(5):839-51. doi: 10.1016/j.jpainsymman.2013.12.240
2. Lee GL, Woo IMH, Goh C. Understanding the concept of a "good death" among bereaved family caregivers of cancer patients in Singapore. *Palliat Support Care* 2013;11(1):37-46. doi: 10.1017/S1478951511000691
3. Hirai K, Miyashita M, Morita T, et al. Good death in Japanese cancer care: a qualitative study. *J Pain Symptom Manage* 2006;31(2):140-47.
4. Sudhakar R, Veeraiah S, Ganesan P, et al. Quality of death: The unspoken experiences of patients with advanced cancers in India - An exploratory qualitative study. *Psychooncology* 2021;30(1):111-17. doi: 10.1002/pon.5570
5. Scarre G. Can there be a good death? *J Eval Clin Pract* 2012;18(5):1082-86. doi: 10.1111/j.1365-2753.2012.01922.x
6. van Lommel P. Near-death experiences: the experience of the self as real and not as an illusion. *Ann N Y Acad Sci* 2011;1234:19-28. doi: 10.1111/j.1749-6632.2011.06080.x
7. Kastbom L, Milberg A, Karlsson M. A good death from the perspective of palliative cancer patients. *Supportive care in cancer : official journal of the Multinational Association of Supportive Care in Cancer* 2017;25(3):933-39. doi: 10.1007/s00520-016-3483-9
8. Mak MHJ. Awareness of dying: an experience of Chinese patients with terminal cancer. *Omega (Westport)* 2001;43(3):259-79.
9. Krikorian A, Maldonado C, Pastrana T. Patient's Perspectives on the Notion of a Good Death: A Systematic Review of the Literature. *J Pain Symptom Manage* 2020;59(1):152-64. doi: 10.1016/j.jpainsymman.2019.07.033
10. Meier EA, Gallegos JV, Thomas LPM, et al. Defining a Good Death (Successful Dying): Literature Review and a Call for Research and Public Dialogue. *Am J Geriatr Psychiatry* 2016;24(4):261-71. doi: 10.1016/j.jagp.2016.01.135
11. Billingham MJ, Billingham S-J. Congruence between preferred and actual place of death according to the presence of malignant or non-malignant disease: a systematic review and meta-analysis. *BMJ Support Palliat Care* 2013;3(2):144-54. doi: 10.1136/bmjspcare-2012-000292
12. Lockwood C, Porritt K, Munn Z, et al. Chapter 2: Systematic reviews of qualitative evidence. In: Aromataris E, Munn Z, eds. JBI Mannual for Evidence Synthesis. Joanna Briggs Institute, 2020:22-44.
13. Butler A, Hall H, Copnell B. A Guide to Writing a Qualitative Systematic Review Protocol to Enhance Evidence-Based Practice in Nursing and Health Care. *Worldviews Evid Based Nurs* 2016;13(3):241-49. doi: 10.1111/wvn.12134
14. Munn Z, Tufanaru C, Aromataris E. JBI's systematic reviews: data extraction and synthesis. *Am J Nurs* 2014;114(7):49-54. doi: 10.1097/01.NAJ.0000451683.66447.89
15. Munn Z, Aromataris E, Tufanaru C, et al. The development of software to support multiple systematic review types: the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). *Int J Evid Based Healthc* 2019;17(1):36-43. doi: 10.1097/XEB.0000000000000152 [published Online First: 2018/09/22]
16. Lockwood C, Munn Z, Porritt K. Qualitative research synthesis: methodological guidance for

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systematic reviewers utilizing meta-aggregation. *International journal of evidence-based healthcare* 2015;13(3):179-87. doi: 10.1097/XEB.0000000000000062

17. Munn Z, Porritt K, Lockwood C, et al. Establishing confidence in the output of qualitative research synthesis: the ConQual approach. *BMC Med Res Methodol* 2014;14:108. doi: 10.1186/1471-2288-14-108

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	7
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7-8
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8-9
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8-9
Study records - data	#11c	Describe planned method of extracting data from reports (such as	9

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collection process		piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	9
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9-10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8-9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

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Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, MENTAL HEALTH, Adult palliative care < PALLIATIVE CARE, QUALITATIVE RESEARCH

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Abstract

Introduction: Promoting the quality of death and achieving a 'good death' is one of the most important goals of palliative care in cancer care. Few studies were conducted to synthesise views of patients with cancer on the notion of a good death and its associated factors. By eliciting the core elements of a good death, the review aims to identify potential unmet needs of patients with cancer and reveal their common values and care preferences at the end of life. The review also has the potential to inform the development of guidelines for clinical care and shared decision making in palliative care practice.

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Ethics and dissemination: Ethical approval is not needed. We will disseminate the findings through international conferences related to cancer care or palliative care. The

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final review will be submitted and published in a peer-reviewed journal.

The Protocol was registered in PROSPERO with the registration number of CRD42021266629.

Keywords: good death, quality of dying, cancer, synthesis, review

Strengths and Limitations of this study

- The qualitative systematic review will be conducted according to the Joanna Briggs Institute methodology for systematic review and synthesis of qualitative data for a high level of rigor.
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Introduction

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Previous research has shown that the concept is complex. Good death means that patients with cancer and stakeholders (i.e., families and caregivers) can be free from distress and suffering and patients are dying in accordance with patients' wishes and clinical, cultural and ethical standards[3]. A qualitative study of patients with cancer, families and healthcare providers in Japan showed that freedom from pain or physical and psychological symptoms is the most frequently cited attribute of a good death, followed by having good family relationships, dying in the preferred place and having good relationships with medical staff[4]. Another study on the bereaved families of patients with cancer reported that physical and psychological suffering, caregivers' distress and emotions are important factors associated with good death[5].

However, good death is dependent on the perspectives of dying individuals and proxy views might not be able to reflect patients' real thoughts[6 7]. A 2007 Sweden qualitative study with 66 adult palliative patients with cancer indicated that patients

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viewed death as a process rather than an event[8]. A good death was deemed a 'successful dying process' when patients are living with the prospect of imminent death, preparing for death and dying comfortably. However, patients in Eastern countries (e.g. China) face challenges when preparing for death because of lower frequencies of bad news (e.g. diagnosis and prognosis) breaking practices and stronger taboos on death discussions compared to those in Western countries. They are more likely to miss the optimal time to discuss the impending death and make preparations for definite departure[9]. Hence, understanding how patients from different cultures view a good death can support development of interventions to through an acculturation approach improve quality of death.

Several reviews have been conducted to enhance understanding of the notion of a good death. However, the findings of the reviews cannot be generalised and used in promoting a good death in cancer because of their limitations. For example, a recent review of the notion of a good death from patients' perspectives[10] found that control of pain and symptom, clear decision-making, feeling of closure, being seen and perceived as a person, prepared for death and being still able to give something to others were core elements for a 'good death'. However, the review included a heterogeneous patient populations, such as patients acquired immune deficiency syndrome, cardiovascular disease, chronic obstructive pulmonary disease and cancer. The sufferings, physical and psychological distress the patients experienced during the cancer trajectory differ from other advanced disease, which might affect their notion on a 'good death'. Hence, the conclusions resulting from a heterogeneous group of patients may not perfectly fit

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patients with cancer. Other reviews included patients with multiple stakeholders, such as families and healthcare providers, or a combination of patients with cancer and non-malignant disease[11 12]. The implications on patients with cancer may be limited. A review of notion of death concentrating to patients' perspective would help elicit and meet their needs in the end of life.

To the best of our knowledge, no review has synthesised qualitative studies to reveal the notion of a 'good death' specifically from the perspective of patients with cancer. By eliciting the core elements of a good death, the review aims to identify potential unmet needs of patients with cancer and reveal their common values and care preferences at the end of life. The review also has the potential to inform the development of guidelines for clinical care and shared decision making in palliative care practice.

Aim

This review aims to identify, appraise and synthesise the existing qualitative evidence exploring the notion of good death of patients with cancer from the perspective of patients. The review will particularly seek to answer the following questions:

1. What are the characteristics of the existing studies that explored a good death for patients with cancer?
2. What are the core elements of a good death for patients with cancer?
3. What are the main factors associated with a good death?

Methods

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Study design

A protocol was developed according to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) and the Joanna Briggs Institute (JBI) methodology for systematic review and synthesis of qualitative data[13]. A meta-aggregative approach will be used to present a comprehensive understanding of patients' perceptions on a 'good death'[14]. The review protocol was submitted to the International prospective register of systematic reviews (PROSPERO) for registration and now is under review.

Inclusion/exclusion criteria

Types of participants: This review will consider studies that only included adult patients with cancer. Studies involving patients with non-cancer diseases, families, healthcare providers, experts or social workers or a mixed group of patients with cancer and other stakeholders (e.g. families, healthcare providers and other stakeholders) will be excluded, because the findings synthesized from a combination group may be unable to reflect the patients' real thought.

Phenomena of interest: This review will consider studies that explore the notion of good death (e.g., the views/perspectives/attitudes/belief about a good death) or main factors (environmental, illness-related or other possible factors) that associated with a good death. Epidemiological studies and studies focusing on certificate of death will be excluded.

Context: This review will include studies conducted in any setting where patients with cancer are cared for, including but not limited to hospitals, hospices, long-term

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care facilities and private residences of patients.

Types of studies: Eligible studies should employ a qualitative methodology, including but not limited to phenomenology, ethnography and grounded theory. Quantitative studies, mixed-method studies, case studies, reviews, commentaries, letters, opinion papers and conference abstracts will be excluded. Only studies published in either English or Chinese language will be included.

Search methods

This review will consider papers published by academic publishers, as well as grey literature. A comprehensive search will firstly be performed on MEDLINE (via PubMed), EMBASE, PsychINFO, CINAHL, Scopus, Web of Science and two Chinese databases (i.e. CNKI and WANGFANG DATA) from inception to October 2021. Grey literature search will then be performed via Google Scholar with the keywords. The initial search strategy is showed in table 1. Reference lists of studies that meet the inclusion criteria will be screened for additional studies.

Table 1 The initial search strategy

‘cancer’ OR ‘neoplasm’
AND
‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’

The initial search strategies were carried out in PubMed in July 2021, and the following queries were used: ("cancer"[Title/Abstract] OR "neoplasm"[Title/Abstract]) AND ("good death"[Title/Abstract] OR "quality of death"[Title/Abstract] OR "quality of dying"[Title/Abstract] OR "good dying"[Title/Abstract]). Two-hundred and sixty-four results were identified. The first author will screen these results firstly with the

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purpose of refining and developing specific search strategies for each database. The full search strategies for all databases can be found in the supplementary file.

Study selection

References identified in the aforementioned databases will be imported into EndNote 20. In the first step, duplicates will be removed with EndNote's deduplicating capabilities, followed by manual screening. JBI System for the Unified Management, Assessment and Review of Information (JBI SUMARI) will be used. JBI SUMARI is a software that supports the entire systematic review process (e.g., screening, quality appraisal, data extraction and data synthesis) for different types of systematic reviews. Second, study screening will be performed, and full texts will be checked according to the inclusion criteria by two reviewers independently. All reasons for exclusion will be recorded, and the results of the screening will be cross-checked between the two reviewers. Disagreements will be resolved through discussion with the third reviewer. In the third step, all the reference lists in the included studies will be checked for additional studies.

Quality appraisal

Included studies will be critically appraised independently using the JBI Critical Appraisal Checklist for Qualitative Research[15]. The instrument consists of 10 items: five items appraising the congruities between research methodology and stated philosophical perspective, research question, data collection methods and data analysis; two items appraising the researchers' roles and cultures; one item appraising the representation of participants and their voices; one item for ethical approval; and one

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item for the interpretation of the data[13]. Two reviewers will be blinded to each other's assessment. The results of the appraisal will only be crosses-checked between the two reviewers after the completion of the initial appraisal of an article. In case of any disagreements, a third reviewer will assist with the appraisal, discussion of the study quality and consensus.

Data extraction

Data extraction will be conducted using the JBI standardised data extraction tool for qualitative research from JBI SUMARI by two reviewers independently. The data to be extracted will include authors and year of publication, journals in which the studies were published, methodologies, methods, phenomena of interest, settings and contexts, characteristics of the participants (age, gender, type of cancer and illness-related information), data analysis, primary findings, authors' conclusions and comments[16]. Any disagreements between the two reviewers will be resolved by the third reviewer.

Data synthesis

Meta-aggregative approach will be adopted for synthesis of the findings. This approach is aligned with the philosophy of pragmatism, suitably answers a specific question and summarises views on good death[16]. First, the findings of the qualitative research will be assembled. Then, the findings will be categorised into groups with different codes according to similarity in meaning. Lastly, the categories will be aggregated for the generation of a set of synthesis findings, which can be used as a basis for evidence-based practice[17]. Only the unequivocal and credible findings will be

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included in the aggregation. Non-supported findings will be presented separately. If the textual pooling is not possible, the findings will be presented in a narrative form.

Confidence in the synthesised findings of the review

This review used the ConQual to rate the confidence of synthesized qualitative findings. In ConQual, dependability and credibility will be determined[18]. The dependability of findings is evaluated using five questions assessing the congruities in the JBI Critical Appraisal Checklist for Qualitative Research. The credibility is established by assessing the congruency between the authors' interpretation and the supporting data. The ConQual will then rank and assign the confidence of the qualitatively synthesised findings into four levels (i.e., High, Moderate, Low or Very low) through comprehensively considering the dependability and credibility assessment. A ConQual summary will be presented along with the type of study, dependency score, credibility score and overall ConQual score.

Ethics and dissemination

Ethical approval is not applicable for this protocol nor the systematic review of qualitative studies. After the completion of the review, the findings will be shared through presentation at international conferences and in an international peer-reviewed journal.

Patient and public Involvement

No patient involved.

Discussion

This will be the first systematic review of qualitative studies that synthesises the

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findings relating to notion of good death specifically from the perspective of patients with cancer. The review is designed to enhance understanding of a good death and has the potential to support the development of an empirical model in promoting a 'good death'. Patients' views on a good death and factors associated with their experiences will be summarized. The new knowledge to be contributed by the review would inform the clinical care and shared-decision making in palliative care. The synthesised findings are expected to have a global impact in the area of cancer care because the review will include patients with different types of cancer and cultural contexts.

Author contributors

XIAO Jinnan designed the study and drafted the manuscript. HUANG Chongmei supervised the study design and critically revised the manuscript. DING Jinfeng contributed to the methods and critically revised the manuscript.

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Competing interest statement

None declared

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References:

1. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin* 2021;71(3):209-49. doi: 10.3322/caac.21660

2. Hales S, Chiu A, Husain A, et al. The quality of dying and death in cancer and its relationship to palliative care and place of death. *J Pain Symptom Manage* 2014;48(5):839-51. doi: 10.1016/j.jpainsymman.2013.12.240

3. Lee GL, Woo IMH, Goh C. Understanding the concept of a "good death" among bereaved family caregivers of cancer patients in Singapore. *Palliat Support Care* 2013;11(1):37-46. doi: 10.1017/S1478951511000691

4. Hirai K, Miyashita M, Morita T, et al. Good death in Japanese cancer care: a qualitative study. *J Pain Symptom Manage* 2006;31(2):140-47.

5. Sudhakar R, Veeraiah S, Ganesan P, et al. Quality of death: The unspoken experiences of patients with advanced cancers in India - An exploratory qualitative study. *Psychooncology* 2021;30(1):111-17. doi: 10.1002/pon.5570

6. Scarre G. Can there be a good death? *J Eval Clin Pract* 2012;18(5):1082-86. doi: 10.1111/j.1365-2753.2012.01922.x

7. van Lommel P. Near-death experiences: the experience of the self as real and not as an illusion. *Ann N Y Acad Sci* 2011;1234:19-28. doi: 10.1111/j.1749-6632.2011.06080.x

8. Kastbom L, Milberg A, Karlsson M. A good death from the perspective of palliative cancer patients. *Support care cancer* 2017;25(3):933-39. doi: 10.1007/s00520-016-3483-9

9. Mak MHJ. Awareness of dying: an experience of Chinese patients with terminal cancer.

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Omega (Westport) 2001;43(3):259-79.

10. Krikorian A, Maldonado C, Pastrana T. Patient's Perspectives on the Notion of a Good Death: A Systematic Review of the Literature. *J Pain Symptom Manage* 2020;59(1):152-64. doi: 10.1016/j.jpainsymman.2019.07.033

11. Meier EA, Gallegos JV, Thomas LPM, et al. Defining a Good Death (Successful Dying): Literature Review and a Call for Research and Public Dialogue. *Am J Geriatr Psychiatry* 2016;24(4):261-71. doi: 10.1016/j.jagp.2016.01.135

12. Billingham MJ, Billingham S-J. Congruence between preferred and actual place of death according to the presence of malignant or non-malignant disease: a systematic review and meta-analysis. *BMJ Support Palliat Care* 2013;3(2):144-54. doi: 10.1136/bmjspcare-2012-000292

13. Lockwood C, Porritt K, Munn Z, et al. Chapter 2: Systematic reviews of qualitative evidence. In: Aromataris E, Munn Z, eds. JBI Mannual for Evidence Synthesis. Joanna Briggs Institute, 2020:22-44.

14. Butler A, Hall H, Copnell B. A Guide to Writing a Qualitative Systematic Review Protocol to Enhance Evidence-Based Practice in Nursing and Health Care. *Worldviews Evid Based Nurs* 2016;13(3):241-49. doi: 10.1111/wvn.12134

15. Munn Z, Aromataris E, Tufanaru C, et al. The development of software to support multiple systematic review types: the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). *Int J Evid Based Healthc* 2019;17(1):36-43. doi: 10.1097/XEB.0000000000000152 [published Online First: 2018/09/22]

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16. Munn Z, Tufanaru C, Aromataris E. JBI's systematic reviews: data extraction and synthesis. *Am J Nurs* 2014;114(7):49-54. doi: 10.1097/01.NAJ.0000451683.66447.89

17. Lockwood C, Munn Z, Porritt K. Qualitative research synthesis: methodological guidance for systematic reviewers utilizing meta-aggregation. *International journal of evidence-based healthcare* 2015;13(3):179-87. doi: 10.1097/XEB.0000000000000062

18. Munn Z, Porritt K, Lockwood C, et al. Establishing confidence in the output of qualitative research synthesis: the ConQual approach. *BMC Med Res Methodol* 2014;14:108. doi: 10.1186/1471-2288-14-108

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Search strategies

1. MEDLINE via PubMed (274)

("cancer"[Title/Abstract] OR "neoplasm"[Title/Abstract]) AND ("good death"[Title/Abstract] OR "quality of death"[Title/Abstract] OR "quality of dying"[Title/Abstract] OR "good dying"[Title/Abstract])

2. EMBASE (391)

(cancer:ti,ab,kw OR neoplasm:ti,ab,kw) AND ('good death':ti,ab,kw OR 'quality of death':ti,ab,kw OR 'quality of dying':ti,ab,kw OR 'good dying':ti,ab,kw)

3. Web of science (639)

1	S1	cancer OR neoplasm [Topic]
2	S2	"good death" OR "quality of death" OR "quality of dying" OR "good dying" [Topic]
3	S1 AND S2 (637)	

Topic: Searches title, abstract, author keywords, and Keywords Plus.

4. PsycINFO via OVID (150)

1	S1	Cancer.mp. or exp neoplasms/
2	S2	Good death.mp.
3	S3	Quality of death.mp.
4	S4	Quality of dying.mp.
5	S5	Good dying.mp.
6	S6	S2 or S3 or S4 or S5 (669)
7	S7	S1 and S6 (150)

5. CINAHL (861)

1	S1	TI (‘cancer’ OR ‘neoplasm’) OR AB (‘cancer’ OR ‘neoplasm’) OR SU (‘cancer’ OR ‘neoplasm’)
2	S2	TI (‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’) OR AB (‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’) OR (‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’)
3	S3	S1 AND S2 (861)

6. Scopus (942)

1	S1	TITLE-ABS-KEY (‘cancer’ OR ‘neoplasm’)
2	S2	TITLE-ABS-KEY (‘good AND death’ OR ‘quality AND of AND death’ OR ‘quality AND of AND dying’ OR ‘good

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		AND dying')
3	S3	S1 AND S2 (942)

Note: ABS: abstract; KEY: key words

7. WANFANG (1521)

1	S1	主题:（'肿瘤' OR 癌）
2	S2	主题:（'好死' OR '歹死' OR '善终' OR '死亡质量' OR '优逝'）
3	S1 AND S2 (1521)	

Limited to Journal articles.

Note: 主题: Topic, including title, abstract and key words.

('肿瘤' OR 癌)=(cancer OR neoplasm).

('好死' OR '歹死' OR '善终' OR '死亡质量' OR '优逝')=('good death' OR 'quality of death' OR 'quality of dying' OR 'good dying')

8. CNKI database

1	S1	主题:（'肿瘤' + 癌）
2	S2	主题:（'好死' + '歹死' + '善终' + '死亡质量' + '优逝'）
3	S1 AND S2 (247)	

Limited to Journal articles.

Note: 主题: Topic, including title, abstract and key words; (肿瘤 OR 癌)=(cancer OR neoplasm); ('好死' OR '歹死' OR '善终' OR '死亡质量' OR '优逝')=('good death' OR 'quality of death' OR 'quality of dying' OR 'good dying'); +: OR

9. Gray literature will be searched through Google and Google Scholar.

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	7
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7-8
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8-9
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8-9
Study records - data	#11c	Describe planned method of extracting data from reports (such as	9

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collection process		piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	9
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9-10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8-9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

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Notion of a good death for patients with cancer: A qualitative systematic review protocol

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Secondary Subject Heading:	Nursing, Qualitative research, Oncology
Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, MENTAL HEALTH, Adult palliative care < PALLIATIVE CARE, QUALITATIVE RESEARCH

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Notion of a good death for patients with cancer: A qualitative systematic review protocol

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Word counts:

1995

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Abstract

Introduction: Promoting the quality of death and achieving a 'good death' is one of the most important goals of palliative care in cancer care. Few studies synthesised views of patients with cancer on the notion of a good death and its associated factors. By eliciting the core elements of a good death, the review aims to identify potential unmet needs of patients with cancer and reveal their common values and care preferences at the end of life. The review also has the potential to inform the development of guidelines for clinical care and shared decision making in palliative care practice.

Methods and analysis: A qualitative systematic review will be conducted and reported according to the Joanna Briggs Institute (JBI) methodology for systematic review and synthesis of qualitative data and the Preferred Reporting Items for Systematic Review and Meta-Analysis. Comprehensive search will be performed in six English and two Chinese databases from inception to 30 April 2022 to retrieve relevant qualitative articles focusing on the notion of a good death from the perspective of patients with cancer. Data will be extracted using the JBI standardised data extraction tool for qualitative research. The quality of the included studies will be critically appraised using the JBI Qualitative Assessment and Review Instrument. Data extraction and quality appraisal will be conducted by two reviewers independently. A meta-aggregative approach and narrative summary will be used to synthesise data.

Ethics and dissemination: Ethical approval is not needed. We will disseminate the findings through international conferences related to cancer care or palliative care. The final review will be submitted and published in a peer-reviewed journal.

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The Protocol was registered in PROSPERO with the registration number of CRD42021266629.

Keywords: good death, quality of dying, cancer, synthesis, review

Strengths and Limitations of this study

- The qualitative systematic review will be conducted according to the Joanna Briggs Institute methodology for systematic review and synthesis of qualitative data to ensure a high level of rigor.
- The review is designed to enhance understanding of a good death and facilitate the building of an empirical intervention model that promotes 'good death'.
- The synthesised findings on good death for patients with cancer are expected to impact on clinical cancer care, shared-decision making and future research in palliative care.
- Some studies can be overlooked because this review will only include studies published in Chinese and English.

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Introduction

The International Agency for Research on Cancer reported 19.3 million new cases of cancer and 10.0 million cancer deaths in 2020 globally[1]. Although the survivorship of a number of cancer types has been substantially prolonged because of the advances in medicine and technology over the recent decades, cancer is still frequently considered being synonymous with the term 'death', especially advanced cancer. A cross-sectional study of 402 cancer deaths in Canada reported that only 39% of deaths were rated 'good' or 'almost perfect'[2]. Promoting the quality of death and achieving a 'good death' is one of the most important goals of palliative care.

Previous research has shown that the concept of good death is complex. It means that patients with cancer and stakeholders (i.e., families and caregivers) can be free from distress and suffering and patients are dying in accordance with patients' wishes and clinical, cultural and ethical standards[3]. A qualitative study of patients with cancer, families and healthcare providers in Japan showed that freedom from pain or physical and psychological symptoms is the most frequently cited attribute of a good death, followed by having good family relationships, dying in the preferred place and having good relationships with medical staff[4]. Another study on the bereaved families of patients with cancer reported that physical and psychological suffering, caregivers' distress and emotions are important factors associated with good death[5].

However, good death is dependent on the perspectives of dying individuals and proxy views might not be able to reflect patients' actual thoughts[6 7]. A 2007 Sweden qualitative study with 66 adult palliative patients with cancer indicated that patients

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viewed death as a process rather than an event[8]. A good death was deemed a 'successful dying process' when patients are living with the prospect of imminent death, preparing for death and dying comfortably. However, patients in Eastern countries (e.g. China) face challenges when preparing for death because of lower frequencies of bad news (e.g. diagnosis and prognosis) breaking practices and stronger taboos on death discussions compared to those in Western countries. They are more likely to miss the optimal time to discuss the impending death and make preparations for definite departure[9]. Hence, understanding how patients from different cultures view a good death can support development of interventions to through an acculturation approach improve quality of death.

Several reviews have been conducted to enhance understanding of the notion of a good death. However, the findings of the reviews cannot be used in promoting a good death in cancer because of their limitations. For example, a recent review of the notion of a good death from patients' perspectives[10] found that control of pain and symptom, clear decision-making, feeling of closure, being seen and perceived as a person, prepared for death and being still able to give something to others were core elements for a 'good death'. However, the review included a heterogeneous patient population, such as patients acquired immune deficiency syndrome, cardiovascular disease, chronic obstructive pulmonary disease and cancer. The sufferings, physical and psychosocial distress the patients experienced during the cancer trajectory differ from other advanced disease, which might affect their notion on a 'good death'. Hence, the conclusions resulting from a heterogeneous group of patients may not perfectly fit patients with

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cancer. Other reviews included patients with multiple stakeholders, such as families and healthcare providers, or a combination of patients with cancer and non-malignant disease[11 12]. The implications on patients with cancer may be limited. A review of notion of death concentrating to patients' perspective would help elicit and meet their needs in the end of life.

To the best of our knowledge, no review has synthesised qualitative studies to reveal the notion of a 'good death' specifically from the perspective of patients with cancer. By eliciting the core elements of a good death, the review aims to identify potential unmet needs of patients with cancer and reveal their common values and care preferences at the end of life. The review also has the potential to inform the development of guidelines for clinical care and shared decision making in palliative care practice.

Aim

This review aims to identify, appraise and synthesise the existing qualitative evidence exploring the notion of good death of patients with cancer from the perspective of patients. The review will particularly seek to answer the following questions:

1. What are the characteristics of the existing studies that explored a good death for patients with cancer?
2. What are the core elements of a good death for patients with cancer?
3. What are the main factors associated with a good death?

Methods

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Study design

A protocol was developed according to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) and the Joanna Briggs Institute (JBI) methodology for systematic review and synthesis of qualitative data[13]. A meta-aggregative approach will be used to present a comprehensive understanding of patients' perceptions on a 'good death'[14]. The review protocol was registered in the international prospective register of systematic reviews (PROSPERO) with the registration number of CRD42021266629.

Inclusion/exclusion criteria

Types of participants: This review will consider studies that only included adult patients with cancer. Studies involving patients with non-cancer diseases, families, healthcare providers, experts or social workers or a mixed group of patients with cancer and other stakeholders (e.g. families, healthcare providers and other stakeholders) will be excluded, because the findings synthesized from a combination group may be unable to reflect the patients' actual thoughts.

Phenomena of interest: This review will consider studies that explore the notion of good death (e.g., the views/perspectives/attitudes/belief about a good death) or main factors (environmental, illness-related or other possible factors) that associated with a good death. Epidemiological studies and studies focusing on certificate of death will be excluded.

Context: This review will include studies conducted in any setting where patients with cancer are cared for, including but not limited to hospitals, hospices, long-term

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care facilities and private residences of patients.

Types of studies: Eligible studies should employ a qualitative methodology, including but not limited to phenomenology, ethnography and grounded theory. Quantitative studies, mixed-method studies, case studies, reviews, commentaries, letters, opinion papers and conference abstracts will be excluded. Only studies published in either English or Chinese language will be included.

Search methods

This review will consider papers published by academic publishers, as well as grey literature. A comprehensive search will firstly be performed on MEDLINE (via PubMed), EMBASE, PsychINFO, CINAHL, Scopus, Web of Science and two Chinese databases (i.e. CNKI and WANGFANG DATA) from inception to October 2021. Grey literature search will be performed via Google Scholar with the same keywords. The initial search strategy is showed in table 1. Reference lists of studies that meet the inclusion criteria will be screened for additional studies.

Table 1 The initial search strategy

‘cancer’ OR ‘neoplasm’
AND
‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’ OR ‘dying well’

The initial search strategies were carried out in PubMed in July 2021, and the following queries were used: ("cancer"[Title/Abstract] OR "neoplasm"[Title/Abstract]) AND ("good death"[Title/Abstract] OR "quality of death"[Title/Abstract] OR "quality of dying"[Title/Abstract] OR "good dying"[Title/Abstract] OR "dying well"[Title/Abstract]). Two-hundred and eight-nine results were identified. The first

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author will screen these results firstly with the purpose of refining and developing specific search strategies for each database. The full search strategies for all databases can be found in the supplementary file.

Study selection

References identified in the aforementioned databases will be imported into EndNote 20. In the first step, duplicates will be removed with EndNote's deduplicating capabilities, followed by manual screening. JBI System for the Unified Management, Assessment and Review of Information (JBI SUMARI) will be used. JBI SUMARI is a software that supports the entire systematic review process (e.g., screening, quality appraisal, data extraction and data synthesis) for different types of systematic reviews. Second, study screening will be performed, and full texts will be checked according to the inclusion criteria by two reviewers independently. All reasons for exclusion will be recorded, and the results of the screening will be cross-checked between the two reviewers. Disagreements will be resolved through discussion with the third reviewer. In the third step, all the reference lists in the included studies will be checked for additional studies.

Quality appraisal

Included studies will be critically appraised independently using the JBI Critical Appraisal Checklist for Qualitative Research[15]. The instrument consists of 10 items: five items appraising the congruities between research methodology and stated philosophical perspective, research question, data collection methods and data analysis; two items appraising the researchers' roles and cultures; one item appraising the

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representation of participants and their voices; one item for ethical approval; and one item for the interpretation of the data[13]. Two reviewers will be blinded to each other's assessment. The results of the appraisal will only be crosses-checked between the two reviewers after the completion of the initial appraisal of an article. In case of any disagreements, a third reviewer will assist with the appraisal, discussion of the study quality and consensus.

Data extraction

Data extraction will be conducted using the JBI standardised data extraction tool for qualitative research from JBI SUMARI by two reviewers independently. The data to be extracted will include authors and year of publication, journals in which the studies were published, methodologies, methods, phenomena of interest, settings and contexts, characteristics of the participants (age, gender, type of cancer and illness-related information), data analysis, primary findings, authors' conclusions and comments[16]. Any disagreements between the two reviewers will be resolved by the third reviewer.

Data synthesis

A meta-aggregative approach will be adopted for synthesis of the findings. This approach is aligned with the philosophy of pragmatism, and suitably answers a specific question and summarises views on good death[16]. First, the findings of the qualitative research will be identified and assembled. Then, the findings will be categorised into groups with different codes according to similarity in meaning. Lastly, the categories will be aggregated for the generation of a set of synthesis findings, which can be used

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as a basis for evidence-based practice[17]. If the textual pooling is not possible, the findings will be presented in a narrative form.

Confidence in the synthesised findings of the review

This review used the ConQual to rate the confidence of synthesized qualitative findings. In ConQual, dependability and credibility will be determined[18]. The dependability of findings is evaluated using five questions assessing the congruities in the JBI Critical Appraisal Checklist for Qualitative Research. The credibility is established by assessing the congruency between the authors' interpretation and the supporting data. The ConQual will then rank and assign the confidence of the qualitatively synthesised findings into four levels (i.e., High, Moderate, Low or Very low) through comprehensively considering the dependability and credibility assessment. A ConQual summary will be presented along with the type of study, dependency score, credibility score and overall ConQual score.

Ethics and dissemination

Ethical approval is not required for this protocol nor the systematic review of qualitative studies. After the completion of the review, the findings will be shared through presentation at international conferences and in an international peer-reviewed journal.

Patient and public involvement

No patient or public involvement.

Discussion

To our knowledge, this will be the first systematic review of qualitative studies that

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synthesises the findings relating to notion of good death specifically from the perspective of patients with cancer. The qualitative systematic review will be conducted according to the JBI methodology for systematic review and synthesis of qualitative data to ensure a high level of rigor. The review is designed to enhance understanding of a good death and has the potential to support the development of an empirical model in promoting a 'good death'. Patients' views on a good death and factors associated with their experiences will be summarized. The new knowledge to be contributed by the review would inform the clinical care and shared-decision making and future research in palliative care.

Author contributors

XIAO Jinnan designed the study and drafted the manuscript. HUANG Chongmei supervised the study design and critically revised the manuscript. DING Jinfeng contributed to the methods and critically revised the manuscript.

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Competing interest statement

None declared

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References:

1. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin* 2021;71(3):209-49. doi: 10.3322/caac.21660

2. Hales S, Chiu A, Husain A, et al. The quality of dying and death in cancer and its relationship to palliative care and place of death. *J Pain Symptom Manage* 2014;48(5):839-51. doi: 10.1016/j.jpainsymman.2013.12.240

3. Lee GL, Woo IMH, Goh C. Understanding the concept of a "good death" among bereaved family caregivers of cancer patients in Singapore. *Palliat Support Care* 2013;11(1):37-46. doi: 10.1017/S1478951511000691

4. Hirai K, Miyashita M, Morita T, et al. Good death in Japanese cancer care: a qualitative study. *J Pain Symptom Manage* 2006;31(2):140-47.

5. Sudhakar R, Veeraiah S, Ganesan P, et al. Quality of death: The unspoken experiences of patients with advanced cancers in India - An exploratory qualitative study. *Psychooncology* 2021;30(1):111-17. doi: 10.1002/pon.5570

6. Scarre G. Can there be a good death? *J Eval Clin Pract* 2012;18(5):1082-86. doi: 10.1111/j.1365-2753.2012.01922.x

7. van Lommel P. Near-death experiences: the experience of the self as real and not as an illusion. *Ann N Y Acad Sci* 2011;1234:19-28. doi: 10.1111/j.1749-6632.2011.06080.x

8. Kastbom L, Milberg A, Karlsson M. A good death from the perspective of palliative cancer patients. *Support care cancer* 2017;25(3):933-39. doi: 10.1007/s00520-016-3483-9

9. Mak MHJ. Awareness of dying: an experience of Chinese patients with terminal cancer.

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Omega (Westport) 2001;43(3):259-79.

10. Krikorian A, Maldonado C, Pastrana T. Patient's Perspectives on the Notion of a Good Death: A Systematic Review of the Literature. *J Pain Symptom Manage* 2020;59(1):152-64. doi: 10.1016/j.jpainsymman.2019.07.033

11. Meier EA, Gallegos JV, Thomas LPM, et al. Defining a Good Death (Successful Dying): Literature Review and a Call for Research and Public Dialogue. *Am J Geriatr Psychiatry* 2016;24(4):261-71. doi: 10.1016/j.jagp.2016.01.135

12. Billingham MJ, Billingham S-J. Congruence between preferred and actual place of death according to the presence of malignant or non-malignant disease: a systematic review and meta-analysis. *BMJ Support Palliat Care* 2013;3(2):144-54. doi: 10.1136/bmjspcare-2012-000292

13. Lockwood C, Porritt K, Munn Z, et al. Chapter 2: Systematic reviews of qualitative evidence. In: Aromataris E, Munn Z, eds. JBI Mannual for Evidence Synthesis. Joanna Briggs Institute, 2020:22-44.

14. Butler A, Hall H, Copnell B. A Guide to Writing a Qualitative Systematic Review Protocol to Enhance Evidence-Based Practice in Nursing and Health Care. *Worldviews Evid Based Nurs* 2016;13(3):241-49. doi: 10.1111/wvn.12134

15. Munn Z, Aromataris E, Tufanaru C, et al. The development of software to support multiple systematic review types: the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). *Int J Evid Based Healthc* 2019;17(1):36-43. doi: 10.1097/XEB.0000000000000152 [published Online First: 2018/09/22]

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16. Munn Z, Tufanaru C, Aromataris E. JBI's systematic reviews: data extraction and synthesis. *Am J Nurs* 2014;114(7):49-54. doi: 10.1097/01.NAJ.0000451683.66447.89

17. Lockwood C, Munn Z, Porritt K. Qualitative research synthesis: methodological guidance for systematic reviewers utilizing meta-aggregation. *International journal of evidence-based healthcare* 2015;13(3):179-87. doi: 10.1097/XEB.0000000000000062

18. Munn Z, Porritt K, Lockwood C, et al. Establishing confidence in the output of qualitative research synthesis: the ConQual approach. *BMC Med Res Methodol* 2014;14:108. doi: 10.1186/1471-2288-14-108

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Search strategies

1. MEDLINE via PubMed

("cancer"[Title/Abstract] OR "neoplasm"[Title/Abstract]) AND ("good death"[Title/Abstract] OR "quality of death"[Title/Abstract] OR "quality of dying"[Title/Abstract] OR "good dying"[Title/Abstract] OR "dying well"[Title/Abstract])

2. EMBASE

(cancer:ti,ab,kw OR neoplasm:ti,ab,kw) AND ('good death':ti,ab,kw OR 'quality of death':ti,ab,kw OR 'quality of dying':ti,ab,kw OR 'good dying':ti,ab,kw OR 'dying well':ti,ab,kw)

3. Web of science

1	S1	cancer OR neoplasm [Topic]
2	S2	"good death" OR "quality of death" OR "quality of dying" OR "good dying" OR "dying well" [Topic]
3	S1 AND S2	

Topic: Searches title, abstract, author keywords, and Keywords Plus.

4. PsycINFO via OVID

1	S1	Cancer.mp. or exp neoplasms/
2	S2	Good death.mp.
3	S3	Quality of death.mp.
4	S4	Quality of dying.mp.
5	S5	Good dying.mp.
6	S6	Dying well.mp.
7	S7	S2 or S3 or S4 or S5 or S 6
8	S8	S1 and S7

5. CINAHL

1	S1	TI (‘cancer’ OR ‘neoplasm’) OR AB (‘cancer’ OR ‘neoplasm’) OR SU (‘cancer’ OR ‘neoplasm’)
2	S2	TI (‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’ OR ‘dying well’) OR AB(‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’ OR ‘dying well’) OR SU (‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’ OR ‘dying well’)
3	S3	S1 AND S2

6. Scopus

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1	S1	TITLE-ABS-KEY (‘cancer’ OR ‘neoplasm’)
2	S2	TITLE-ABS-KEY (‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’ OR ‘dying well’)
3	S3	S1 AND S2

Note: ABS: abstract; KEY: key words

7. WANFANG

1	S1	主题:（‘肿瘤’ OR 癌）
2	S2	主题:（‘好死’ OR ‘歹死’ OR ‘善终’ OR ‘死亡质量’ OR ‘优逝’）
3	S1 AND S2	

Limited to Journal articles.

Note: 主题: Topic, including title, abstract and key words.

(‘肿瘤’ OR 癌)=(cancer OR neoplasm).

(‘好死’ OR ‘歹死’ OR ‘善终’ OR ‘死亡质量’ OR ‘优逝’)=(‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’)

8. CNKI database

1	S1	主题:（‘肿瘤’+ 癌）
2	S2	主题:（‘好死’+‘歹死’+‘善终’+‘死亡质量’+‘优逝’）
3	S1 AND S2	

Limited to Journal articles.

Note: 主题: Topic, including title, abstract and key words; (肿瘤 OR 癌)=(cancer OR neoplasm); (‘好死’ OR ‘歹死’ OR ‘善终’ OR ‘死亡质量’ OR ‘优逝’)=(‘good death’ OR ‘quality of death’ OR ‘quality of dying’ OR ‘good dying’); +: OR

9. Gray literature will be searched through Google and Google Scholar with the same keywords.

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	7
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7-8
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8-9
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8-9
Study records - data	#11c	Describe planned method of extracting data from reports (such as	9

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collection process		piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	9
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9-10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8-9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

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